Spending a Fixed Compute Budget: Agents, Rounds, Reasoning Depth, and Sampling

Study `openrouter_agent_allocation_v1`

17 September 2026

Abstract

A fixed inference budget can be spent in several ways. It can buy more agents working in parallel, more rounds of revision, a longer reasoning allowance inside a single call, or simply more independent samples read out by a vote. This report asks which of those purchases is worth making, holding total compute exactly constant.

We define the budget as $C = B \cdot R \cdot A$, the product of the number of agents, the number of rounds, and the per-agent per-round reasoning allowance. The study evaluated 95 configurations on one fixed 400-item panel spanning four benchmarks, with 67,514 graded generations and a 25-sample independent bank per item that makes oracle pass@k available at every budget. Because C is materialised per configuration, allocations that differ in shape but agree in cost can be compared directly. At $C = 8A$, for instance, seven distinct allocations were run.

Three findings hold across benchmarks. First, at matched compute the shape of the allocation barely matters: of 70 per-benchmark comparisons against independent sampling at the same budget, none survives multiplicity correction, and the largest effect anywhere is 6 accuracy points. Second, every allocation plateaus near what two samples and a vote already deliver, while oracle pass@k continues to climb, so the headroom that compute makes available but no configuration collects grows from zero to 6.7 accuracy points on the three-benchmark macro and to 9 points on MMLU-Pro. Third, letting agents see each other's work actively destroys accuracy, and the damage grows with rounds: at five agents and four peer rounds the penalty is 5.0 points on the macro and 10 points on MBPP+.

The binding constraint in this regime is not generation but selection. Going from one sample to twenty-five moves 16 points of MMLU-Pro items out of the state where no correct answer exists, but only 1.1 of those points end up in a state a vote can deliver; the other 15 land in a state where the correct answer is present and outvoted.

We then ask which axis would collapse that gap. Ranking the levers on the same sealed data, a reranker with pairwise AUC 0.70 applied to candidates the run already paid for beats every one of the 95 allocations, and AUC 0.80 delivers 0.876 against an oracle of 0.917; today's plurality vote is equivalent to AUC 0.63. A larger per-draw reasoning allowance is the only generation-side lever that helps, halving the MATH-500 gap. More agents and more rounds contribute close to nothing. The prescription is to spend on verification, not on allocation.

1 Methods

1.1 Model, benchmarks and panels

Generations come from `google/gemini-2.5-flash-lite` through OpenRouter with the provider pinned, fallbacks disabled and caching off, at temperature 0.7 with nucleus sampling at 0.95. A replication arm uses `google/gemini-2.5-flash`.

Four benchmark families are scored: MATH-500 with `math_verify`, MMLU-Pro with exact option match, a four-subtask zero-shot BBH subset (logical deduction, shuffled objects, boolean expressions, temporal sequences) with exact task match, and MBPP+ graded by the full EvalPlus suite. BBH is reported as this four-subtask subset, not the official full BBH score, and its aggregate weights the four subtasks equally.

The main panel is 400 items, 100 per benchmark. A high-allowance panel of 96 items and a replication panel of 48 items are nested subsets of it, so every cross-panel comparison uses shared item ids only. All 95 main-panel configurations share one fixed item population, which is what allows every contrast in this report to be paired item by item.

1.2 The compute accounting

Each configuration is a triple (B, R, A) and the budget is their product. The sealed tables carry B, R and A as explicit columns, so no parsing of configuration names is involved:

Family	Notation	B	R	C
Independent samples, vote	`sample/k`	k	1	kA_0
One agent revising	`private_single/t`	1	t	tA_0
Team, no peer exposure	`private_team/n/r`	n	$1+r$	$n(1+r)A_0$
Team, peer exposure	`peer/n/r`	n	$1+r$	$n(1+r)A_0$
One deeper call	`deep/m`	1	1	mA_0

A_0 is 1024 tokens on the main panel and 4096 on the high panel. The `deep` family scales A rather than the call count, so its budget must be read from A and not from the number of generations; a build-time check asserts that the derived C equals the sealed requested reasoning tokens for every configuration on every panel, and that $B \cdot R$ equals the sealed generation count everywhere except `deep`, where it is one call by construction.

Figure 1 shows the resulting design. Fourteen budget levels carry two or more distinct allocations, and the levels $C/A_0 \in \{4, 8\}$ carry seven each.

1.3 What counts as a result, and what does not

A configuration delivers exactly one answer per item. Text benchmarks select it by plurality vote; MBPP+ selects by prompt-visible public tests only, and is therefore reported as `public_select@k` and never as a vote. Delivered accuracy counts operational failures, unparseable outputs and abstentions as incorrect, so nothing is silently dropped.

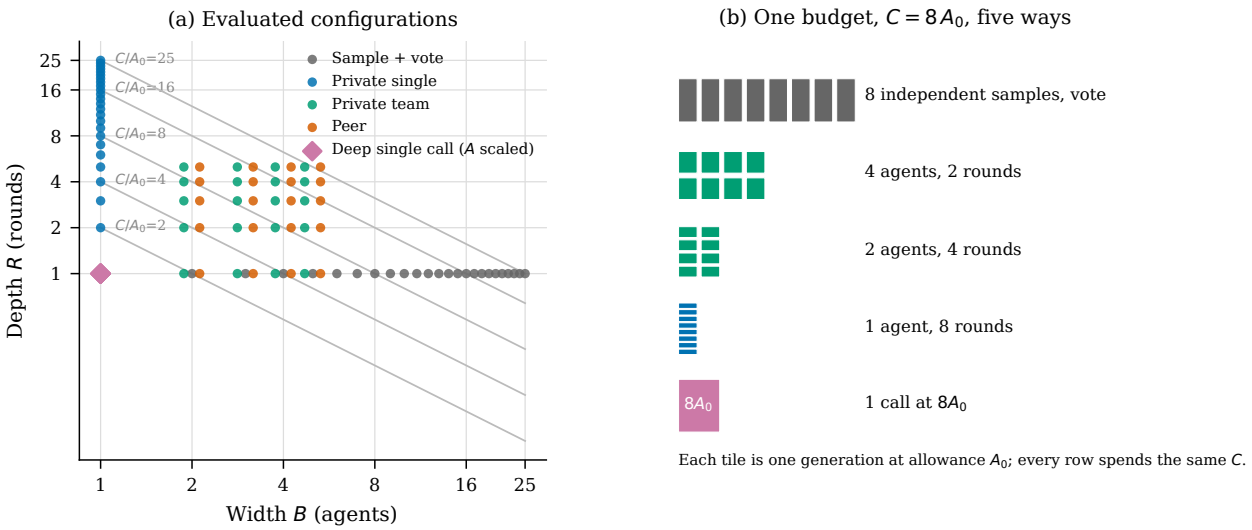


Figure 1: The design. (a) Every evaluated configuration in the width-depth plane; grey lines are iso-compute contours, along which every point spends the same C. Private-team and peer points are nudged apart because they occupy identical (B, R) cells and differ only in whether agents see each other. (b) The five ways one budget of $8A_0$ was actually spent.

Oracle pass@k is not an available selector. It is reported as a ceiling only, using the standard combinatorial estimator $1 - \binom{n-c}{k} / \binom{n}{k}$ over the 25-sample bank. Two normalised quantities are used throughout:

Amplification is delivered accuracy minus the accuracy of independent sampling at the same budget, in accuracy points, paired per item. It answers whether an allocation beats simply drawing $k = C/A_0$ samples and reading them out.

Ceiling capture is $\eta = (\text{accuracy} - \text{pass@1})/(\text{coverage@}k - \text{pass@1})$, the share of the headroom that the budget makes available which the configuration actually delivers. The denominator is the realised prefix coverage of the same k samples, which is binary per item and therefore pairs with the numerator under the same resample indices. A value of 1 means the configuration collected everything the budget put within reach; 0 means it did no better than one sample. Values above 1 are possible and meaningful, because a revising or communicating configuration can produce an answer that independent sampling never generated. We suppress η where the denominator falls below 0.02, which masks BBH entirely.

1.4 Estimation and multiplicity

All intervals are percentile intervals from a 10,000-replicate item-clustered bootstrap, stratified by benchmark and with BBH subtasks weighted equally. Every configuration shares the same resample indices, so differences between configurations are correctly paired.

Two contrasts were prespecified and frozen before collection: peer versus private team at $B{=}4$, $r{=}4$, and the communicate-then-private decomposition at $h{=}3$. Those carry Holm-corrected p-values and are quoted directly from the sealed contrast table. Everything else in this report is exploratory. In the amplification matrix we mark cells whose nominal paired interval excludes zero, and we also report how many survive Benjamini-Hochberg control within each panel, which turns out to be the more informative number.

Macro averages are reported two ways: over all four benchmarks, which reconciles with the run report, and

over the three informative ones. BBH is saturated at 1.000 in every configuration, so including it inflates the level and compresses every contrast toward zero.

1.5 Limitations

A is a requested allowance, not realised reasoning; realised token counts are recorded separately and are consistently below the request. This is a token-accounted and dollar-accounted study, not a FLOP-matched one. With 100 items per benchmark, one item is one accuracy point, so differences below roughly 3 points are not separable from noise. Results are from a single small model, and the peer-communication finding in particular should not be assumed to transfer to larger models. The text completion headroom was amended twice during piloting, from 1024 to 2048 to 4096 tokens, and the main run was collected at 4096 under a clarified system prompt. Frontiers shown here are descriptive: the best among evaluated configurations on this sample, never a validated policy.

2 Results

2.1 Allocation shape barely matters; the ceiling keeps rising

Figure 2 is the headline. On every benchmark the oracle ceiling climbs steadily with compute while every delivered curve flattens almost immediately.

On the three-benchmark macro a single sample delivers 0.780 and two samples with a vote deliver 0.823. Twenty-five samples deliver 0.833, an additional 1.0 point for twelve times the compute. The best configuration anywhere in the design is a five-agent private team with three revision rounds at 0.850, which is 2.7 points above sample-and-vote at a comparable budget and 7.0 points above one sample. Over the same range oracle pass@k rises from 0.780 to 0.917.

Per benchmark the picture differs in an instructive way. On MATH-500 plain sampling is essentially unbeatable: it reaches 0.960, which is the best value any configuration attains, while the oracle reaches 1.000. On MMLU-Pro allocation helps most, with a three-agent private team at one revision round reaching 0.780 against 0.720 for two samples. On MBPP+ the achievable readout actually *declines* with budget, from 0.810 at one sample to 0.790 at twenty-five, while oracle coverage rises to 0.890: selecting code by prompt-visible tests gets worse as the candidate pool grows, because a candidate that passes the visible assertion and fails the hidden suite becomes easier to find.

Benchmark	pass@1	vote@2	vote@25	best any	best archive	oracle@25	headroom
MATH-500	0.890	0.940	0.960	0.960	0.960	1.000	0.040
MMLU-Pro	0.640	0.720	0.750	0.780	0.780	0.860	0.090
BBH-logic4	1.000	1.000	1.000	1.000	1.000	1.000	0.000
MBPP+	0.810	0.810	0.790	0.820	0.854	0.890	0.070
Macro (3)	0.780	0.823	0.833	0.850	0.861	0.917	0.067

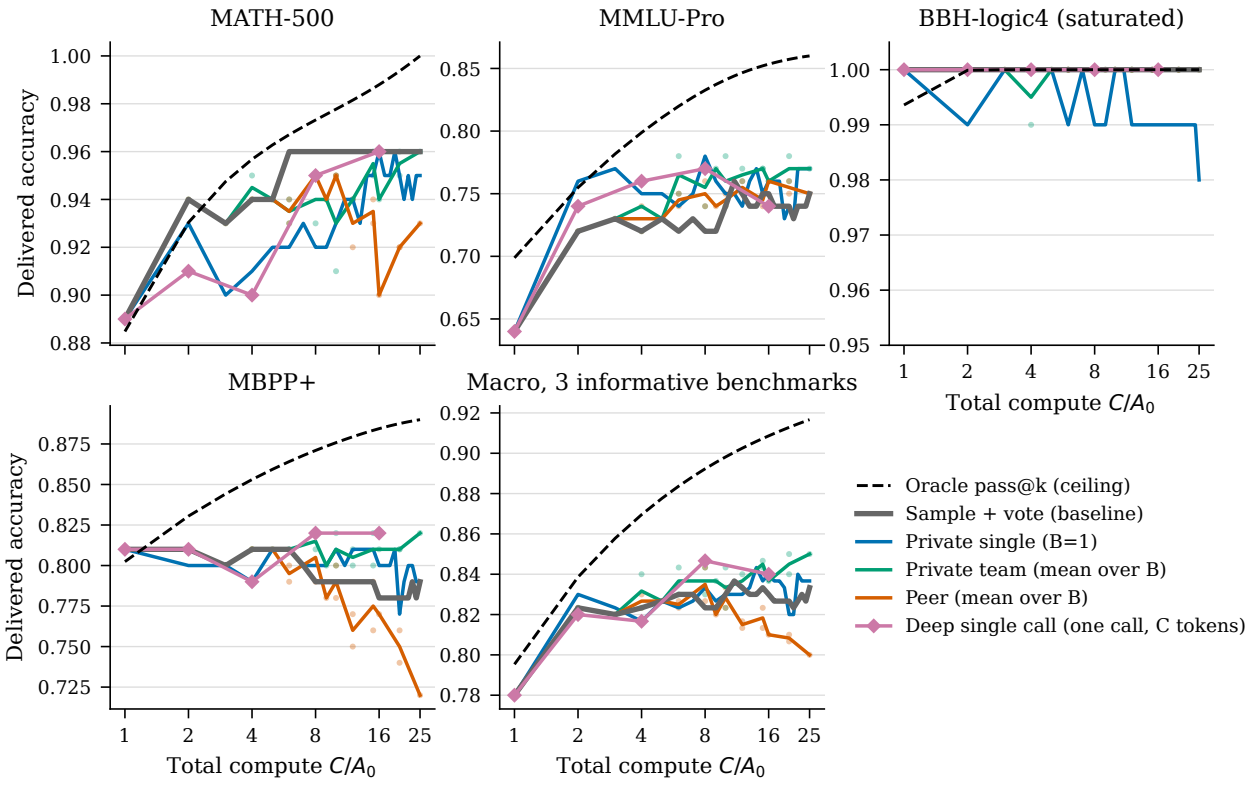


Figure 2: Delivered accuracy against total compute. All curves at a given x spend the same budget. The dashed ceiling is oracle pass@k, an unavailable selector. Faint points are individual team sizes; family lines are the mean over B. BBH is drawn on a fixed window because it is saturated.

2.2 The amplification matrix

Figure 3 places every allocation against independent sampling at the same budget.

The matrix is mostly pale, and that is the result. Across 70 comparisons per benchmark, zero survive multiplicity correction; the smallest p-value on the macro is 0.017, far above the Benjamini-Hochberg threshold. Only 2 macro cells and 8 MATH-500 cells clear even an uncorrected interval. A practitioner choosing among these allocations at fixed cost is, on this evidence, choosing among options that cannot be reliably distinguished.

The signs are nonetheless consistent and worth reading. Private teams sit on the positive side on MMLU-Pro (up to $+6$ points) and MBPP+ (up to $+3$), and on the negative side on MATH-500. Peer configurations sit on the negative side almost everywhere and grow more negative toward the bottom of the matrix, that is, at larger budgets with more rounds. The `deep` column is positive on MBPP+ ($+3$ to $+4$) and negative on MATH-500.

Figure 4 normalises the same comparison against the headroom available at each budget.

Capture falls as budget rises. On the macro the median η is 0.67 at the smallest budgets and 0.39 by $C/A_0 \geq 16$; on MMLU-Pro it falls from 0.80 to 0.50. On MBPP+ the median capture at large budgets is -0.25, meaning the typical configuration there ends up below a single sample even as coverage improves. The one cell above 1.0, a single agent revising once on MMLU-Pro, is genuine: revision produced correct answers that neither independent sample contained.

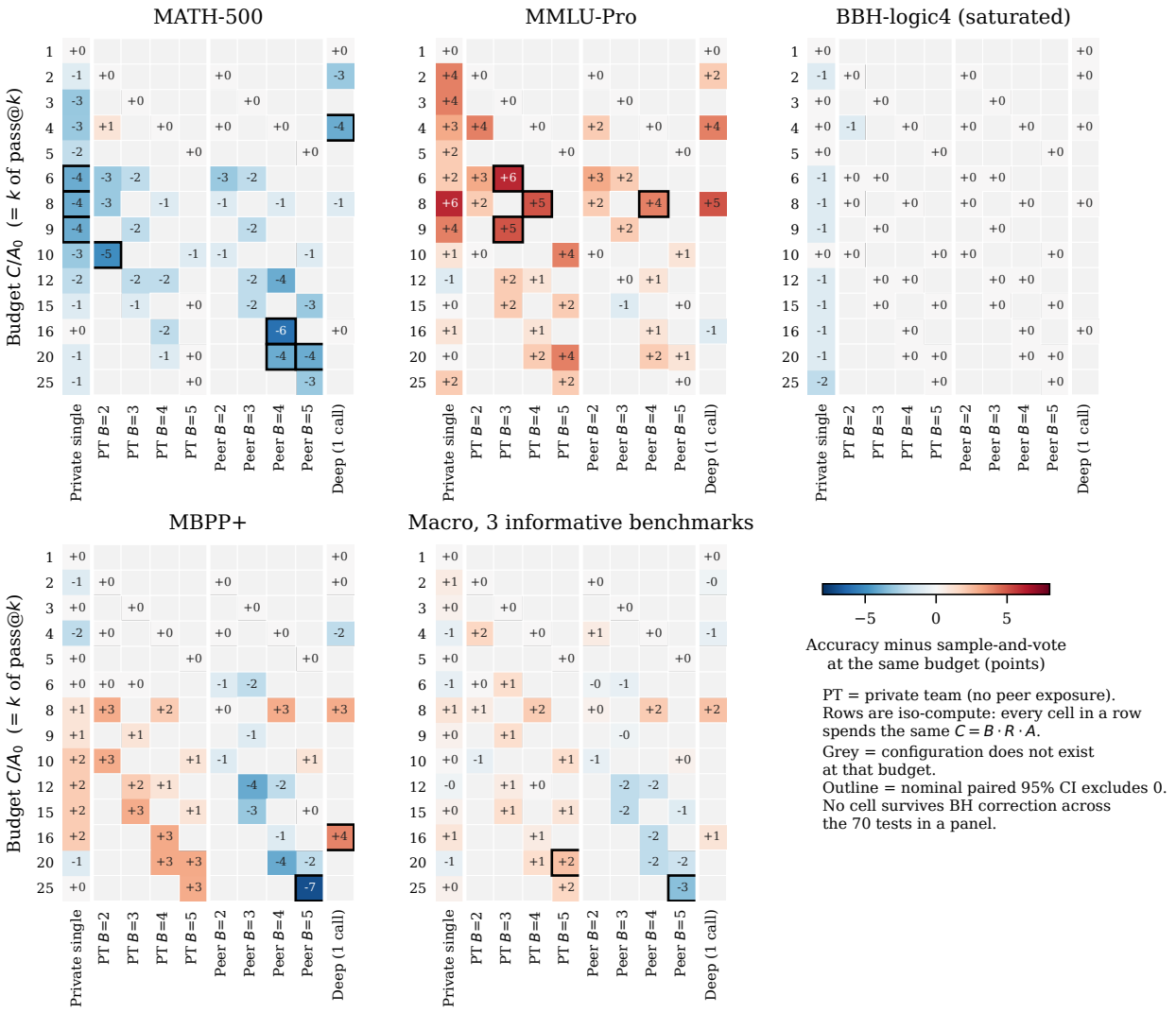


Figure 3: Amplification relative to sampling. Rows are iso-compute budgets; every cell in a row spends the same C. Values are delivered accuracy minus sample-and-vote at that budget, in accuracy points. Outlines mark nominal paired intervals excluding zero; *no* cell survives Benjamini-Hochberg control across the 70 tests in a panel. PT is a private team, that is, a team without peer exposure.

2.3 Width against depth

Figure 5 isolates the shape question by fixing the budget and varying only how it is split.

For private teams, wider is mildly better than deeper: on the macro the mean amplification rises from about 0 at B=1 to +1.6 points at B=5, and the same upward slope appears on MATH-500, MMLU-Pro and MBPP+. Pure depth, one agent revising many times, is the weakest private allocation on every informative benchmark. For peer configurations the slope is absent or reversed and the whole curve sits below zero on the macro and on MBPP+.

Two readings follow. Spreading a budget across a few agents is slightly better than spending it on one agent thinking for longer in sequence, but the advantage is within noise. And the gap between the private and peer curves, which is the pure effect of letting agents see each other, is larger and more consistent than the

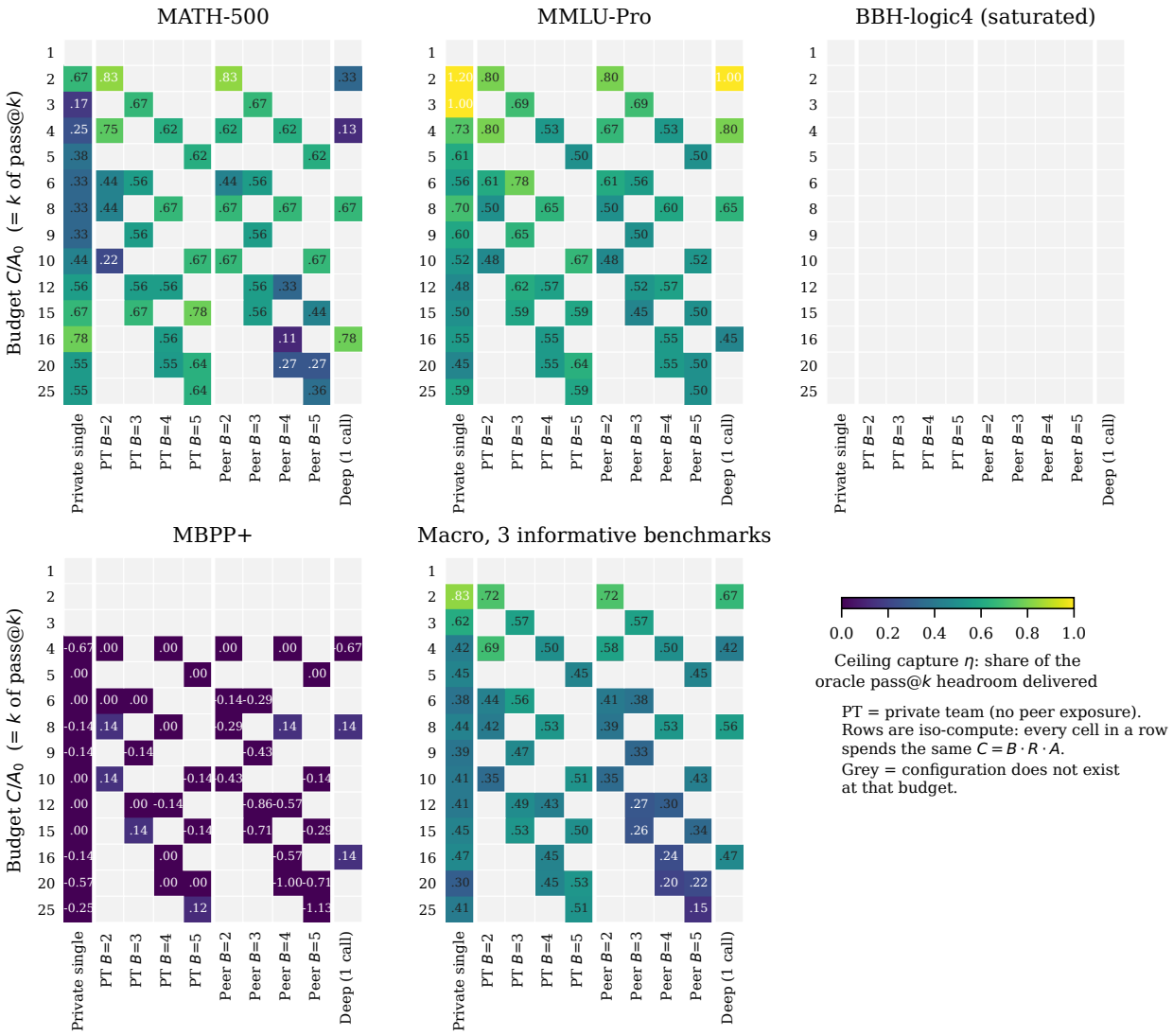


Figure 4: Ceiling capture η: the share of the headroom a budget makes available that the configuration actually delivers. BBH is masked because its denominator is zero.

width-depth effect it is plotted against.

2.4 Generation is not the bottleneck

Figure 6 makes the central claim directly.

At the largest budget the best configuration on the macro reaches 0.850 against an oracle of 0.917, leaving 6.7 points unrealised. Per benchmark the unrealised headroom is 4.0 points on MATH-500, 9.0 on MMLU-Pro and 7.0 on MBPP+. The band widens monotonically: at $C = 2A_0$ it is near zero, and every additional unit of compute after that adds more to the ceiling than to what any policy delivers.

The blue line is deliberately generous. It takes the maximum over up to seven configurations at each budget, so it is biased upward by selection on this very sample, and the gap is still large.

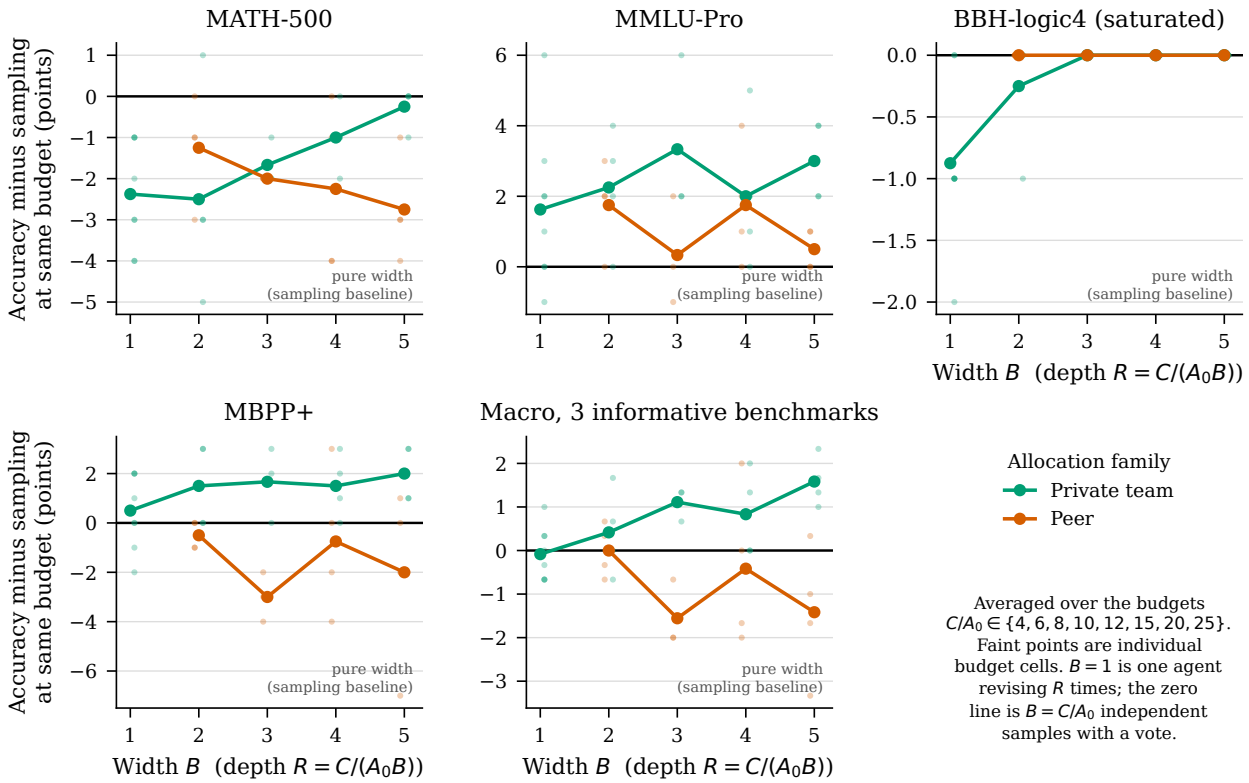


Figure 5: Allocation shape at fixed compute, averaged over budgets $C/A_0 \in \{4, 6, 8, 10, 12, 15, 20, 25\}$. B=1 is one agent revising R times. The zero line is the same budget spent on $B = C/A_0$ independent samples.

2.5 The cost of letting agents talk

The one prespecified contrast that resolves is the comparison between teams that see each other and teams that do not, at matched size and matched rounds.

Contrast	Level	n	Estimate	95% CI	Holm p
Peer − private team, B=4, r=4	Macro (4)	400	-0.0250	[-0.046, -0.008]	0.027
	MATH-500	100	-0.0300	[-0.064, -0.003]	n/a
	MMLU-Pro	100	+0.0000	[+0.000, +0.000]	n/a
	BBH-logic4	100	+0.0000	[+0.000, +0.000]	n/a
	MBPP+	100	-0.0700	[-0.144, -0.010]	n/a
CP − private-team gain, h=3	Macro (4)	192	-0.0052	[-0.025, +0.009]	0.567
	MATH-500	48	-0.0208	[-0.021, +0.004]	n/a
	MMLU-Pro	48	+0.0000	[+0.000, +0.000]	n/a
	BBH-logic4	48	+0.0000	[+0.000, +0.000]	n/a
	MBPP+	48	+0.0000	[-0.053, +0.053]	n/a

Peer exposure at four agents and four rounds costs 2.50 accuracy points on the four-benchmark macro, with a 95% interval of $[-0.046, -0.008]$ and a Holm-corrected p of 0.027. The effect is carried by MBPP+ (-7.0 points) and MATH-500 (-3.0); it is exactly zero on MMLU-Pro and on saturated BBH. The second prespecified contrast, the communicate-then-private decomposition at h=3, is null at -0.005 with p =

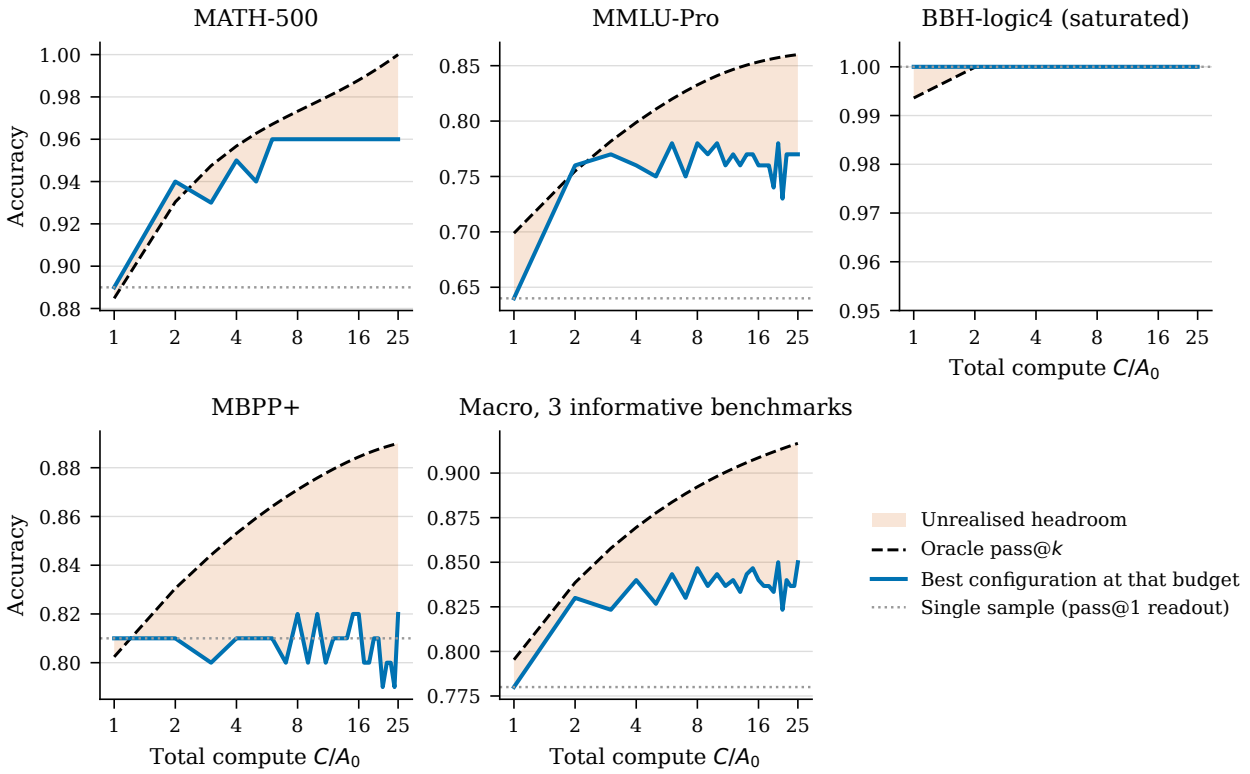


Figure 6: The selection gap. The shaded band is headroom that compute made available and no evaluated configuration collected. The blue line is the maximum over all configurations at that budget, so it is an optimistic envelope and still falls short.

0.567.

Figure 7 shows that this is not a single unlucky cell but a trend.

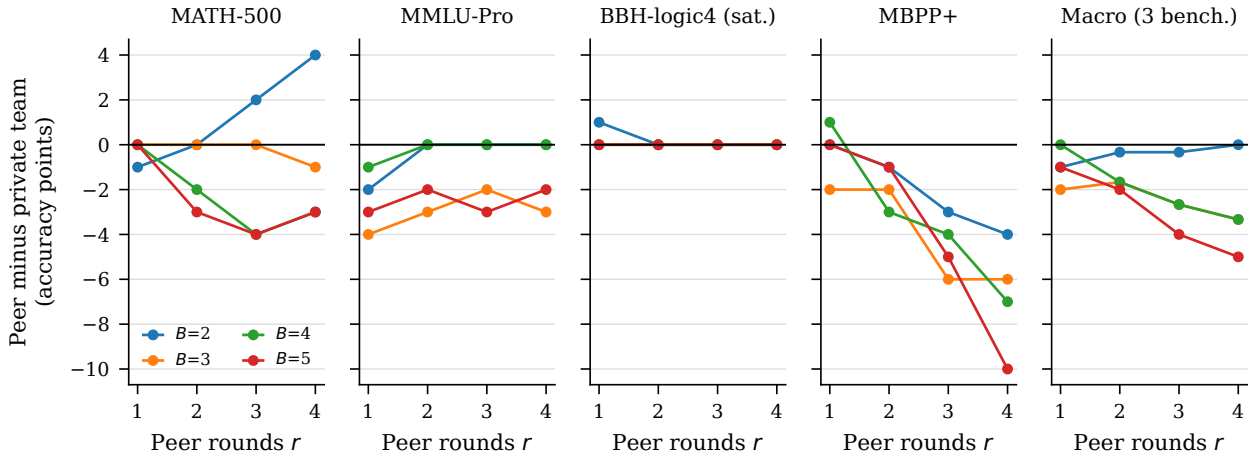


Figure 7: Peer minus private team at matched team size and matched rounds. The only difference between the two arms is whether agents see each other's drafts.

The penalty deepens with rounds on every informative benchmark and with team size on MBPP+. At the largest configuration, five agents and four peer rounds, the cost is 5.0 points on the macro ($[-0.077, -0.023]$) and 10.0 points on MBPP+ ($[-0.17, -0.04]$).

Figure 8 shows the mechanism. Revision rounds move a substantial number of answers in both directions, and the net is close to zero.

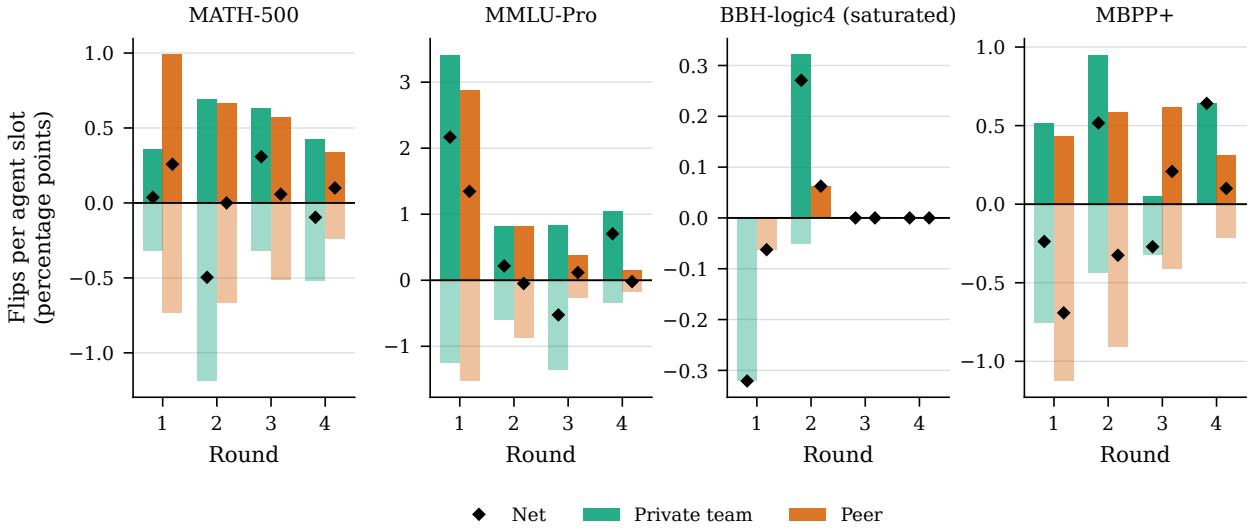


Figure 8: Churn per revision round. Solid bars are wrong-to-correct flips, faded bars are correct-to-wrong, and the diamond is the net. Rates are per agent slot, in percentage points.

On MMLU-Pro the first round has a clearly positive net and later rounds do not, which is why one revision round is the best MMLU-Pro configuration in the whole design. On MATH-500 and MBPP+ the net is negative for several rounds. Peer exposure raises the correct-to-wrong rate more than it raises the wrong-to-correct rate, which is exactly the shape of a consensus process that propagates confident errors.

2.6 Depth inside one call

The `deep` arm spends the whole budget on a single call with a larger reasoning allowance, so it pays no repeated prompt cost and has no opportunity to select.

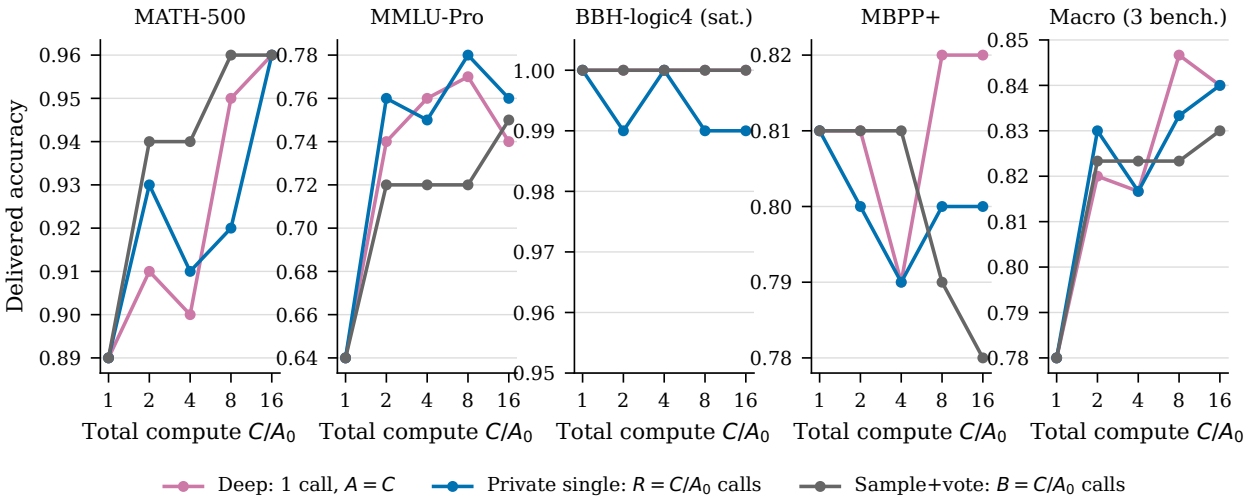


Figure 9: The same budget as one long call, as sequential revisions, or as parallel samples.

One deeper call is competitive with everything else and is the single best configuration on MBPP+ (0.820

at $C = 8A_0$, against 0.790 for eight samples). On the macro it reaches 0.847, within 0.3 points of the best configuration in the entire design, using one generation rather than twenty. On MATH-500 it trails sampling slightly. Since it requires no orchestration, no repeated context and no selector, it is the most practical of the allocations tested.

2.7 What this costs in dollars

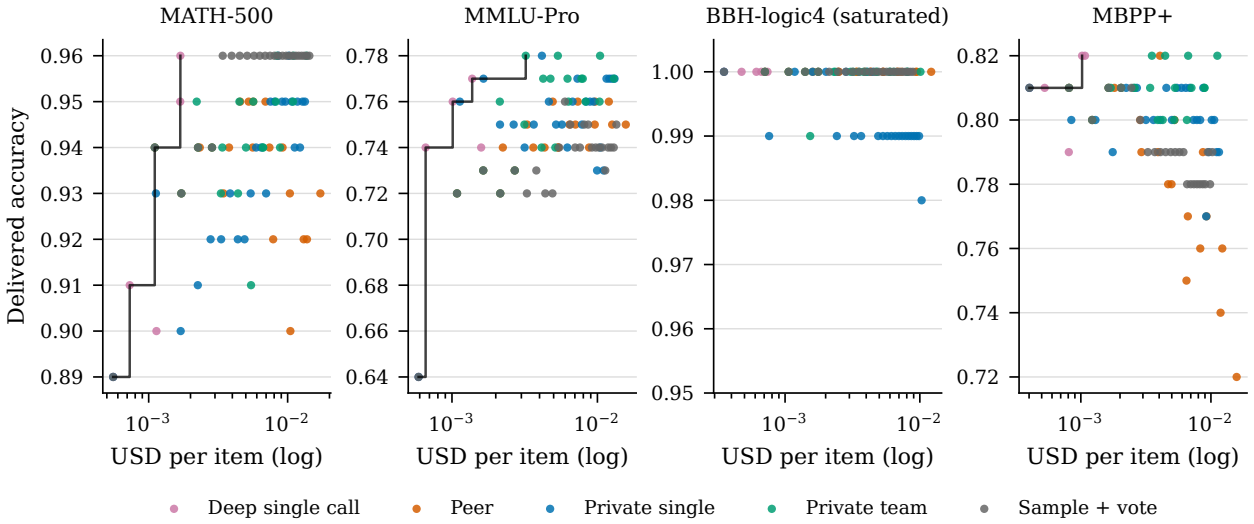


Figure 10: Delivered accuracy against realised spend per item. The step line is the descriptive Pareto front over evaluated configurations.

The frontier saturates almost immediately. One sample costs \$0.000475 per item and one deeper call at $8A_0$ costs \$0.001207; five agents over four peer rounds cost \$0.015196, which is 32 times the single sample, and deliver less. Everything beyond the deeper single call at \$0.001207 per item is, on this evidence, spent without return.

2.8 Robustness

Raising the allowance to $A_0 = 4096$ lifts the whole picture, with pass@1 at 0.833 and the best configuration reaching 0.903, but the qualitative shape survives: the ceiling is 0.931 and 2.8 points remain unrealised. The Gemini 2.5 Flash replication behaves the same way, with a best configuration of 0.806 against a ceiling of 0.833 at $k = 16$. Both panels are small, 96 and 48 items, so these are consistency checks rather than independent confirmations. The direction of the peer penalty is preserved on both.

2.9 Anatomy of the gap

Every configuration satisfies, per item, *delivered = pool coverage − selection loss*, where coverage is 1 when the configuration's own candidate pool contained a correct answer. Comparing that coverage with independent sampling at the same budget splits the gap in two: a generation-side part, where the pool never held the answer, and a selection-side part, where it did and the selector missed it.

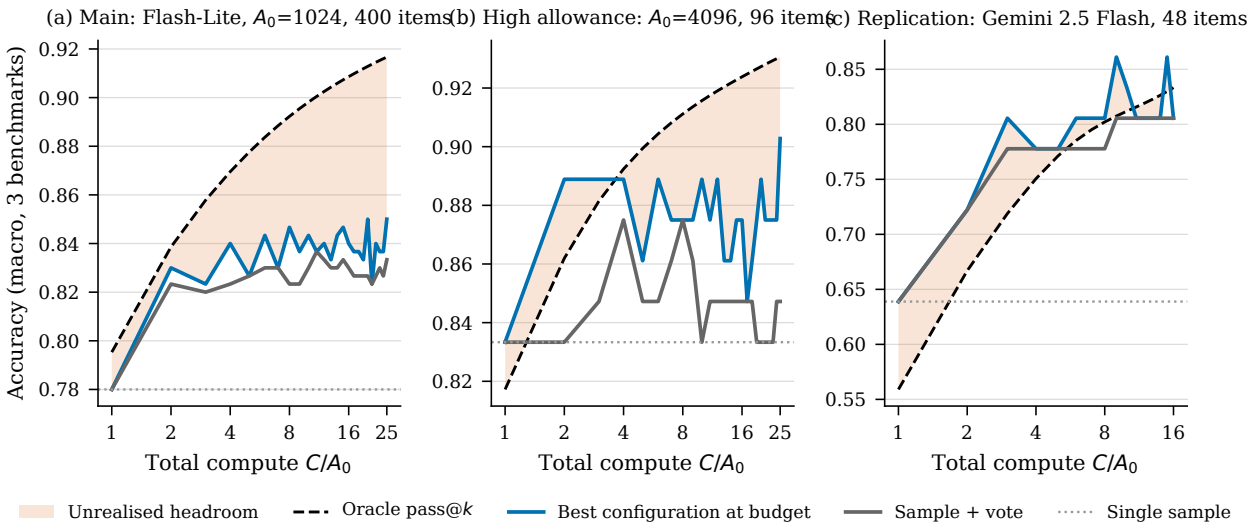


Figure 11: The same comparison at four times the reasoning allowance and on a stronger model, on shared items only. The 96-item and 48-item panels have unstable intervals.

Allocation	Pool coverage	Delivered	Coverage deficit	Selection loss	Archive
`private_team/5/4`	0.903	0.850	0.013	0.053	0.850
`private_single/25`	0.837	0.837	0.080	0.000	0.840
`sample/25`	0.917	0.833	0.000	0.083	0.833
`peer/5/4`	0.807	0.800	0.110	0.007	0.837
Sampling ceiling	0.917				

The split is stark. Independent sampling has *zero* coverage deficit by construction and its entire 8.3-point gap is selection loss. One agent revising 25 times has the opposite profile: essentially no selection loss, because its delivered pool is a single final answer, and an 8.0-point coverage deficit. Peer teams are worst on coverage, at 11.0 points, because mutual exposure collapses the pool onto a shared answer. No allocation in the study is generation-limited in the sense that matters: at every budget, some allocation holds the correct answer.

Figure 13 explains why coverage is nearly worthless on its own. Given that a correct answer is in the pool, the probability of delivering it is 0.305 on MATH-500, 0.20 on MMLU-Pro and 0.06 on MBPP+ when correct answers make up under a fifth of the pool, and rises to essentially 1 on the text benchmarks once they pass half. A vote does not need the correct answer to be present, it needs it to be modal.

That is what the stacked panels show. On MMLU-Pro, going from one sample to twenty-five reduces the share of items with no correct answer anywhere from 0.301 to 0.140. Almost none of that 16.1 points becomes deliverable: the majority share rises only from 0.699 to 0.710, and 15.0 points land in the middle band where a correct answer exists and loses the vote. Extra draws convert impossible items into unusable ones. The middle band is the oracle gap, exactly.

The archive readout is the one free lever. Voting over every round's output rather than the last restores `private_team/5/4` coverage to 0.917, precisely the 25-sample ceiling, at no extra tokens. As a delivered gain it is worth $+0.037$ on `peer/5/4` (95% CI $[0.017, 0.060]$) and $+0.033$ on `peer/4/4`, but only 0.000 on `private_team/5/4`. It repairs what peer exposure discards; it does not lift the frontier.

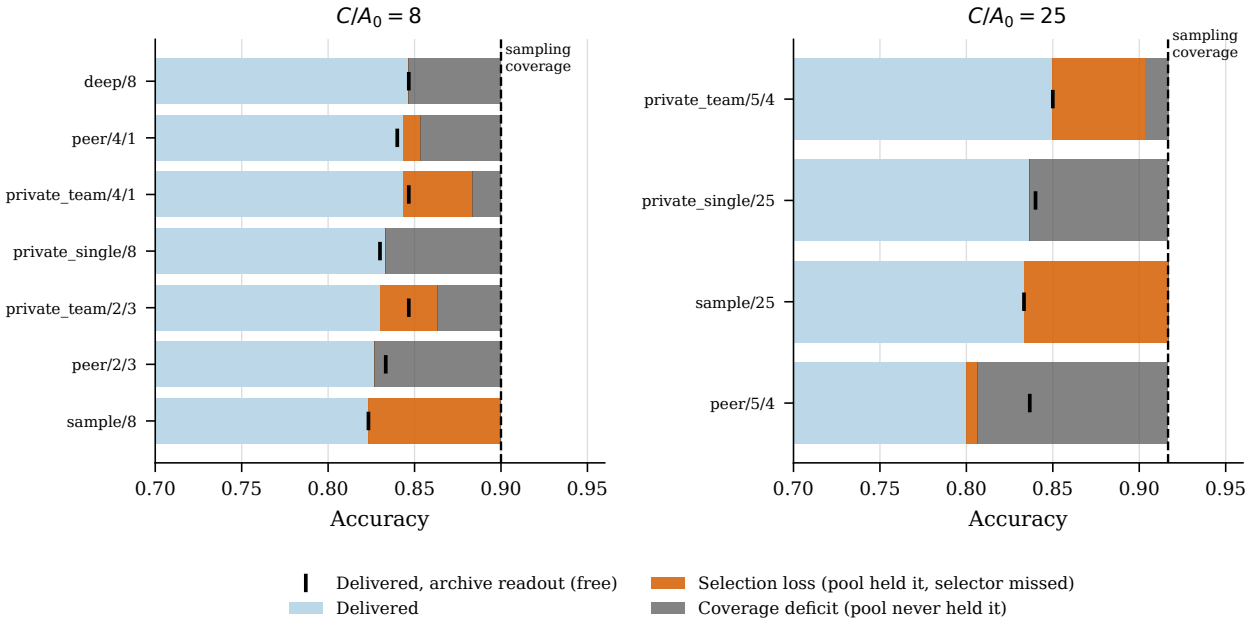


Figure 12: Each allocation's budget-matched gap, split into the two losses. Bars sum to the coverage that independent sampling achieves at the same budget. Tick marks show the archive readout, which votes over every round's output instead of only the last and costs nothing extra.

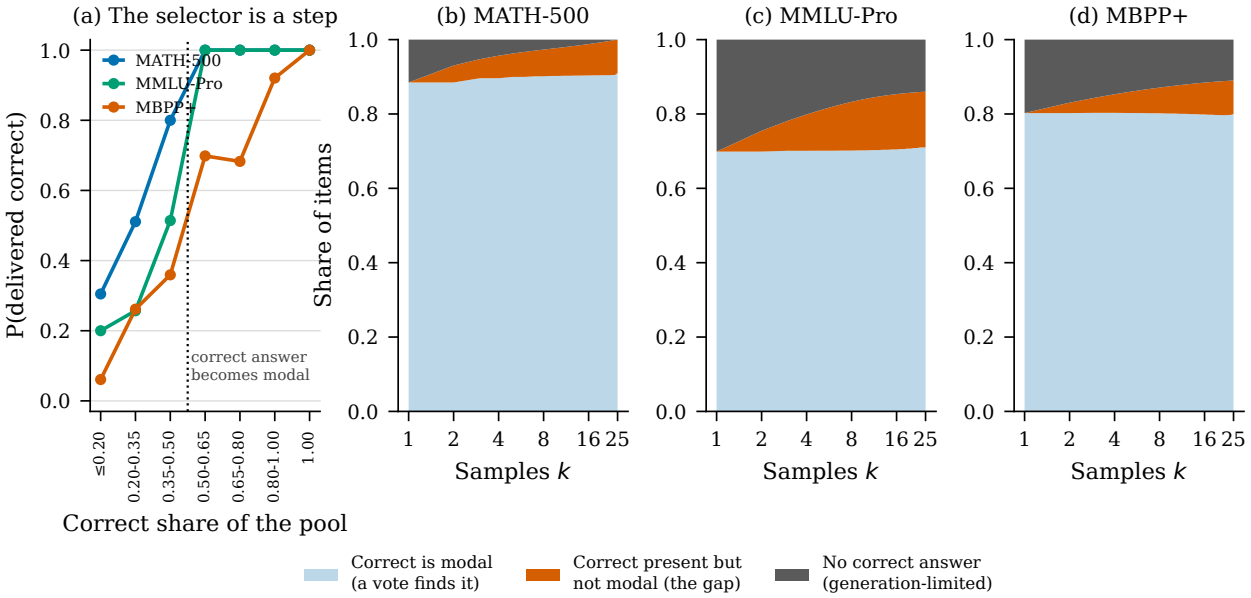


Figure 13: (a) Probability of delivering a correct answer given that one is in the pool, against the correct share of that pool. (b to d) Where the item mass sits for a k-sample prefix, computed exactly from the bank by the hypergeometric prefix distribution, with ties broken as half.

2.10 What a verifier would buy

If selection is the constraint, the natural question is how good a selector would have to be. Figure 14 answers it on the real pool compositions by scoring correct candidates $\mathcal{N}(\mu, 1)$ and incorrect ones $\mathcal{N}(0, 1)$ and taking the argmax, which makes the verifier's pairwise AUC $\Phi(\mu/\sqrt{2})$.

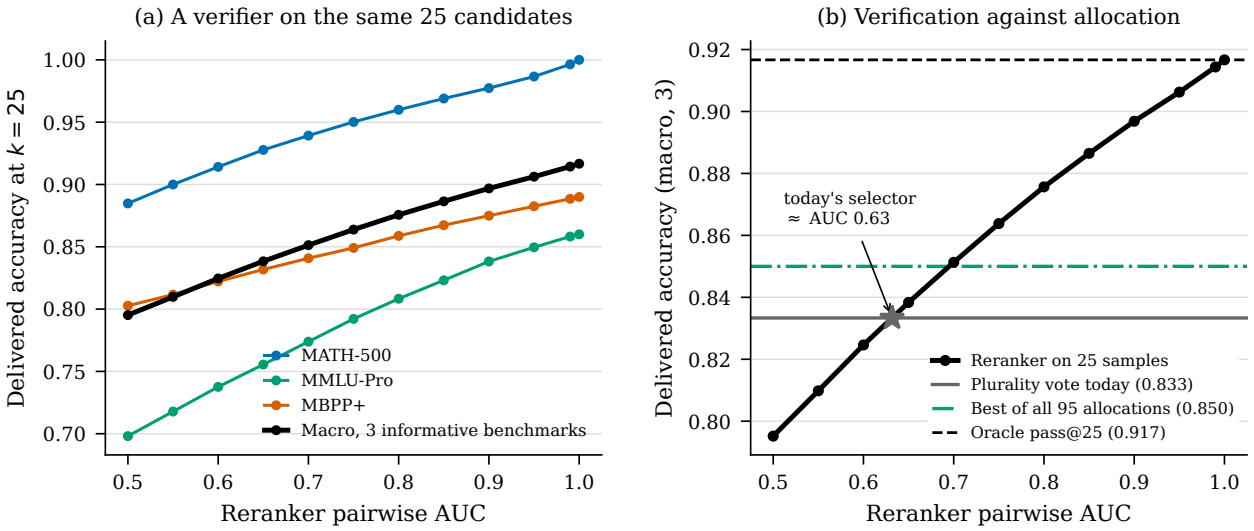


Figure 14: Delivered accuracy at $k = 25$ from a reranker of given pairwise AUC, applied to candidates the run already generated. This bounds what discriminative power is worth; it is not evidence that such a verifier exists for these tasks.

Today's plurality vote delivers 0.833 on the macro, which is what a reranker of AUC 0.63 would achieve. A reranker at AUC 0.70 delivers 0.851, already above the best of all 95 allocations (0.850). At AUC 0.80 it delivers 0.876, which is 2.6 points beyond anything the allocation design reached, and at AUC 0.90 it delivers 0.897 against an oracle of 0.917.

The comparison is the point. The entire design, 95 configurations and 67,514 generations spanning every combination of agents, rounds and reasoning allowance, moves delivered accuracy over a range of about 7 points. A mediocre verifier applied to twenty-five samples already paid for moves it further.

2.11 The one generation-side lever, and one selector that backfires

Because the gap is governed by per-draw accuracy rather than draw count, the one axis that should help is A. On the 96 shared items, raising the allowance from 1024 to 4096 lifts MATH-500 per-draw accuracy from 0.830 to 0.902 and halves the gap from 0.083 to 0.042. On MMLU-Pro per-draw accuracy rises from 0.717 to 0.753 and the gap is unchanged at 0.083. On MBPP+ per-draw accuracy falls slightly and the gap widens from 0.083 to 0.125. A helps exactly where longer reasoning raises per-draw accuracy, and not otherwise.

Among allocations, the only lever that moves the relevant quantity is concentration. At $C = 25A_0$ private teams raise the correct share of the pool from 0.795 under sampling to 0.843, which is why they deliver 0.850 against 0.833. Peer exposure lowers it to 0.797. Private teams help not because they explore more but because they consolidate, which is what a vote can use.

Finally, one defect worth fixing on its own. MBPP+ selects by prompt-visible tests, and from $k = 8$ onward that selector performs *below* a random draw: 0.790 against 0.802 at $k = 25$, having started at 0.810 for a single candidate. Enlarging a code candidate pool and filtering on visible assertions finds programs that

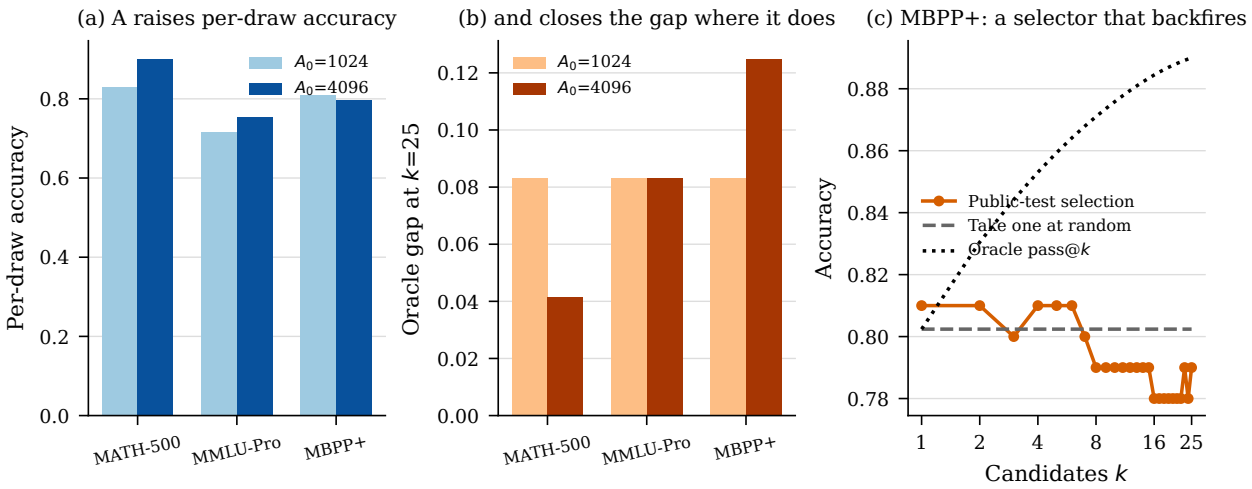


Figure 15: (a, b) The reasoning allowance on the 96 shared items. (c) MBPP+ public-test selection against simply taking one candidate at random.

satisfy the visible evidence and fail the hidden suite. On this benchmark the run would have done better selecting at random, and better still doing nothing.

2.12 Does the allocation change how well a pool reads out?

Delivering a correct answer requires the pool to hold one, so for every configuration *delivered* = *coverage* $\times S$ exactly, where S is selection success, the chance of delivering given the pool held a correct answer. That identity turns the allocation question into a question about which of the two factors an axis buys.

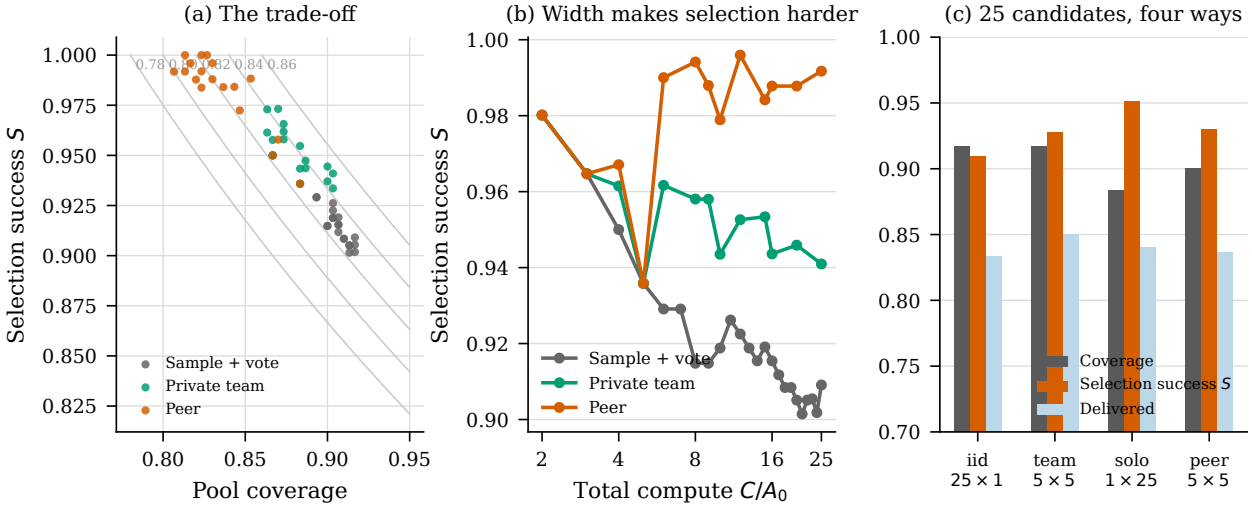


Figure 16: (a) Every configuration at $C/A_0 \geq 4$ in the coverage-selectability plane; grey lines are iso-delivered contours, since the product of the axes is delivered accuracy. Private single is absent because its current-view pool is one candidate, so it has no selection to measure. (b) Selection success against budget. (c) Four ways to produce 25 candidates.

The answer is that allocation changes selection success a great deal, and in the direction opposite to its effect on coverage. Scaling horizontally drives S down: independent sampling falls from 0.965 at the smallest

budgets to 0.905 at the largest, because each extra candidate is another way for the vote to fragment. Scaling vertically with peer exposure drives S up, from 0.970 to 0.989, because the agents converge on one answer and the vote becomes trivial. Private teams sit between, at 0.967 falling to 0.944.

Neither extreme wins, because each pays for its factor with the other. Peer configurations have the best selection success in the entire study and the worst coverage; independent sampling has the best coverage and the worst selection success. Panel (a) shows the consequence: the configurations lie along an anti-diagonal band and therefore mostly on the same iso-delivered contour. This is the mechanism behind the report's first finding. Allocation shape is close to irrelevant not because the axes do nothing, but because what they give on one factor they take back on the other.

The controlled comparison is panel (c), where pool size is held at 25 and only the way those 25 candidates were produced varies. Twenty-five independent samples and a five-agent, five-round team read over its archive have *identical* coverage, 0.917. The team's pool is more concentrated, 0.829 against 0.795, and its selection success is higher, 0.927 against 0.909, so it delivers 0.850 against 0.833. The difference is $+0.017$ with a 95% interval of $[-0.003, 0.037]$, so the mechanism is measured cleanly while the delivered advantage is not separable from noise at 400 items. One agent revising 25 times has the highest selection success of all, 0.951, and loses on coverage, at 0.883.

The practical reading is that a mix is the only allocation that can hold both factors at once: parallel agents give coverage, revision gives concentration, and reading out over the archive rather than the final round keeps both. That is a reason to prefer it, but this study does not have the power to prove it pays.

2.13 Repairing the selector

If selection is the constraint, the structure of the answer space decides which repair is even available. Figure 17 shows that this structure differs enormously across benchmarks.

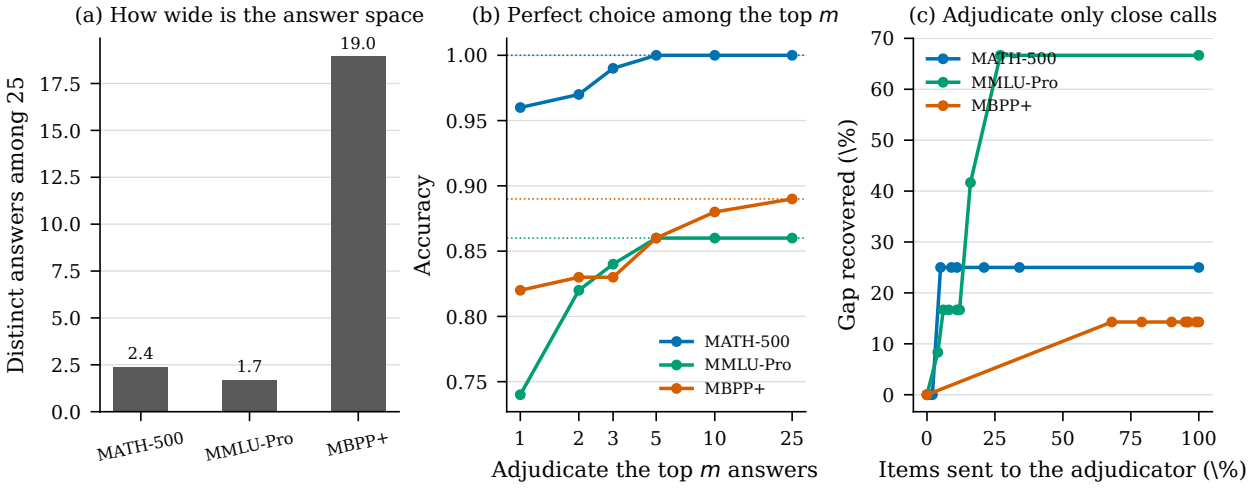


Figure 17: (a) Distinct answers among 25 samples. (b) Accuracy from a perfect choice restricted to the m most-voted distinct answers; dotted lines are each benchmark's oracle. (c) Gap recovered against the share of items sent to an adjudicator, gating on the plurality margin.

Twenty-five MMLU-Pro samples produce only 1.7 distinct answers, with the top one taking 0.891 of the vote. MATH-500 produces 2.4. MBPP+ produces 19.0, with the top program taking 0.170, because programs are nearly all textually unique. These are three different selection problems wearing the same name.

Where the answer space is narrow, the repair is cheap. On MMLU-Pro a perfect choice between just the *top two* voted answers lifts accuracy from 0.740 to 0.820, which is $+0.080$ with a 95% interval of $[0.030, 0.140]$ and recovers two thirds of the 0.120 gap. By the top five it reaches 0.860, the full oracle. On the twelve items where a correct answer exists and the vote misses it, the correct answer is ranked second in two thirds of cases and ranked 2.58 on average. So the verifier this benchmark needs is not a reranker over 25 candidates; it is one binary comparison per item.

Where the answer space is wide, adjudication does almost nothing. On MBPP+ the correct program sits at rank 7.43 among the failing items, top-two adjudication is worth $+0.010$ $[0.000, 0.030]$, and recovering the gap requires ranking essentially the whole pool. Code needs a verifier that scores each candidate on its own, which in practice means better tests rather than better voting. That is consistent with the public-test selector failing here: it is the only benchmark whose selector already tries to score candidates individually, and it does so on evidence the candidates were optimised against.

Gating helps less than it should. Sending only the closest 16% of MMLU-Pro items to an adjudicator recovers 0.417 of the gap, and recovering the full 0.667 requires sending all of them. The reason is visible in panel (a): the mean plurality margin on MMLU-Pro is 0.813, so most failures are not close calls. They are items where the model is confidently and consistently wrong, and an adjudicator has to be willing to overrule a large majority. That is a harder verifier to build than a tie-breaker, and it is the central difficulty this study surfaces.

2.14 A conditional scaling law

The sections above say what the gap is and what closes it on average. This one asks the sharper question: given a budget and a problem, which axis should take the budget? The answer turns out to follow from a small amount of probability, and the derivation is worth doing carefully because it says exactly when each axis stops working.

Setup

Fix an item. A single generation at allowance A draws an answer from a distribution over answer classes, where a class is a set of answers the selector treats as the same. Write

- p for the probability that a draw lands in the correct class,
- q_1 for the probability of the largest single *incorrect* class,
- $m = p - q_1$ for the **plurality margin**.

The pair (p, q_1) is the item's state. Both are properties of the model, the prompt and the allowance A; neither depends on how many samples we draw. Drawing k independent samples and taking the plurality gives counts $(N_c, N_w, \dots)$ that are multinomial with k trials.

Two remarks on scope. The selector must actually be a vote, so this subsection covers MATH-500 and MMLU-Pro; MBPP+ selects by public tests and is treated separately below. And p is the mass of the correct *class*, which matters because this run's math selector merges semantically equivalent answers.

The limit: what a vote can ever do

Proposition 1. *As $k \to \infty$, the plurality is correct with probability* $\mathbf{1}[m > 0]$.

By the law of large numbers $N_c/k \to p$ and $N_w/k \to q_1$ almost surely, so the correct class outnumbers every incorrect class eventually if and only if $p > q_1$. Sampling more is a consistent estimator of the model's own modal answer, and nothing more.

This is the whole ceiling result in one line. Oracle pass@k asks whether a correct answer appeared at all, which happens with probability $1 - (1-p)^k \to 1$ for any $p > 0$. A vote asks whether the correct answer appeared *most often*, which converges to a hard zero or one. The gap between them, in the limit, is exactly the probability that $0 < p \le q_1$: the item where the model can produce the right answer but not usually.

On the two vote benchmarks, $\mathbf{1}[m > 0]$ predicts the realised outcome at $k = 25$ on 0.990 of items. Across all three informative benchmarks, including MBPP+ where the selector is not a vote, the agreement is 0.968.

The rate: how fast a vote gets there

Proposition 2. *For k draws,*

$$P(\text{plurality correct}) \;\approx\; \Phi\left(\frac{m\sqrt{k}}{\sigma}\right), \qquad \sigma^2 \;=\; p + q_1 - m^2.$$

Let $D_k = N_c - N_w$. Under the multinomial, $\mathbb{E}[D_k] = k(p - q_1) = km$, and since $\mathrm{Cov}(N_c, N_w) = -k\,p\,q_1$,

$$\mathrm{Var}(D_k) = kp(1-p) + kq_1(1-q_1) + 2kpq_1 = k\big(p + q_1 - (p - q_1)^2\big) = k\sigma^2.$$

The plurality is correct when $D_k > 0$, and the central limit theorem gives $P(D_k > 0) \approx \Phi(km/\sqrt{k\sigma^2}) = \Phi(m\sqrt{k}/\sigma)$.

Note there is nothing to fit. Every quantity is read off the item's state. Figure 18(a) tests it against simulated plurality outcomes on 1800 item-by-k points across 200 items: the correlation is 0.996, the mean absolute error 0.007, and the bias -0.004.

Figure 18(b) draws the resulting family. Three features matter. The curves are concave in k, so each extra sample is worth less than the last. They saturate, so past a point extra samples are worth nothing at all. And for $m < 0$ the curve *declines*: sampling harder makes the answer worse, converging on the wrong class.

How many samples a decided item needs

Corollary. *To be correct with probability* $1 - \alpha$ *requires*

$$k^*(m) \;=\; \left(\frac{z_{1-\alpha}\,\sigma}{m}\right)^2 \;=\; \frac{z_{1-\alpha}^2\,(p + q_1 - m^2)}{m^2}.$$

Inverting Proposition 2 gives $m\sqrt{k}/\sigma \ge z_{1-\alpha}$. When two classes carry the mass, $p + q_1 = 1$ and this simplifies to $k^* = z^2(1 - m^2)/m^2$, which behaves as z^2/m^2 for small margins. **Width buys resolution at a rate quadratic in the inverse margin.** At $\alpha = 0.05$:

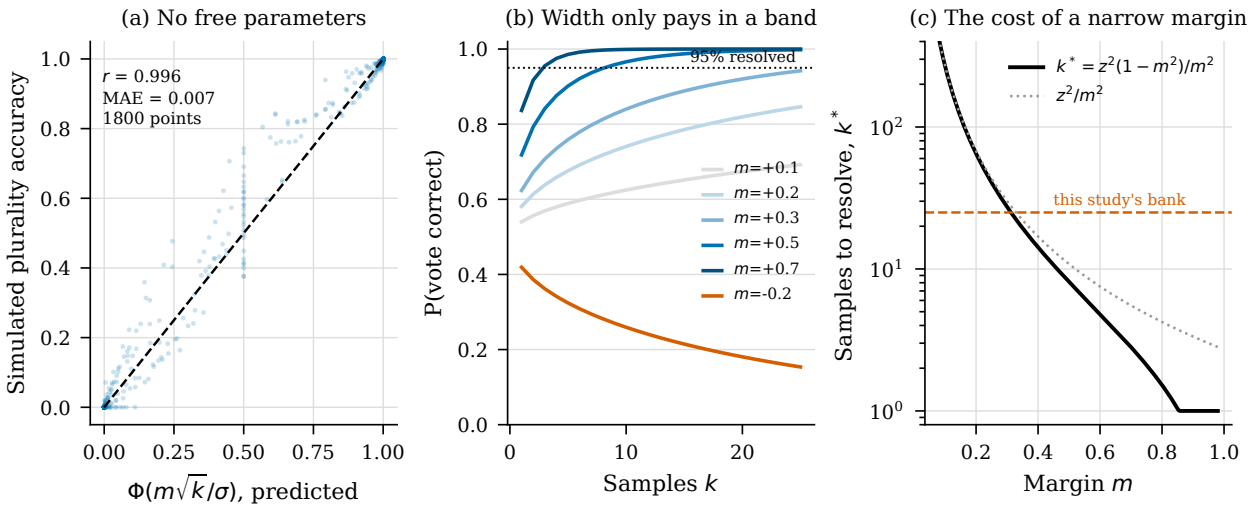


Figure 18: (a) The parameter-free prediction against simulated plurality accuracy, one point per item and k. (b) The curves the model implies: sub-linear gains, a hard ceiling, and a declining curve once $m < 0$. (c) The closed form for k^*, with this study's bank size marked.

margin m	0.1	0.2	0.3	0.5	0.7	0.9
k^*	268	65	27	8.1	2.8	1

This is the mechanism behind the plateau in Figure 2. Most decided items have wide margins, so k^* is one or two and every sample after that is spent re-answering a settled question. Meanwhile the items that would need many samples need more than the bank holds: a margin of 0.2 wants 65 draws and the study bought 25. Width is useful only in a narrow band of margins, roughly $0.3 \lesssim m \lesssim 0.8$ at this bank size, and few items live there. A direct log-log fit of the observed resolution point against margin across all three benchmarks gives $k^* = 1.04\, m^{-2.17}$ with $\mathrm{corr}(\log m, \log k^*) = -0.875$, consistent with the predicted exponent of -2; the fit is noisier than the theory because k^* is floored at one sample and measured on a coarse grid.

Three regimes, and what each axis does to them

The sign of m and whether p is zero partition every item into three regimes, drawn in Figure 19:

Regime	Condition	What compute does
Decided	$m > 0$	A vote converges to the right answer in $k^*(m)$ samples. Beyond that, every axis is equivalent and every extra token is waste.
Outvoted	$p > 0,\ m \leq 0$	A vote converges to the *wrong* answer. Width is negative. Only a change to (p, q_1) helps.
Dead	$p = 0$	Nothing in the sampling distribution is correct. Only a change to the distribution helps.

Now the axes. **Width** B increases k and leaves (p, q_1) untouched, so by Proposition 1 it cannot change which class wins; it only reduces the noise around a predetermined answer. This is why width is bounded by

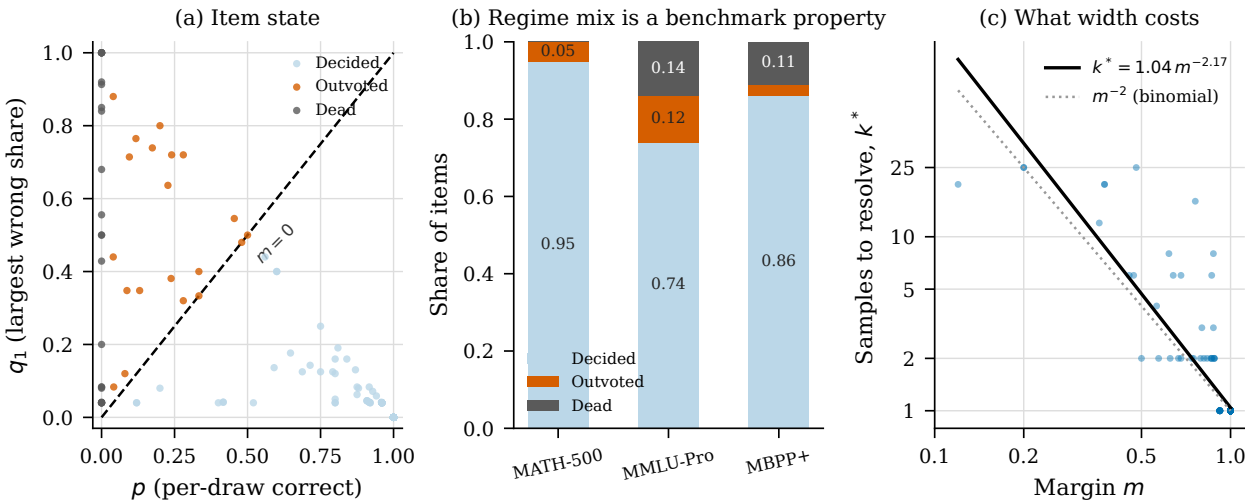


Figure 19: (a) Every item in the state plane; the dashed diagonal is $m = 0$, so points above it are outvoted and points on the left edge are dead. (b) The regime mix is a stable property of the benchmark. (c) Samples needed to bring a decided item within 5% of its limit, against its margin, with the fitted law and the binomial prediction.

k^* on decided items and actively harmful on outvoted ones. **Depth in one call**, raising A, changes the sampling distribution itself and is therefore the only axis in this study that can move an item between regimes. **Sequential revision** also changes the distribution but, as Section 2.12 showed, mostly by concentrating it, raising the correct share of the pool from 0.795 to 0.843 without adding coverage.

The empirical check is that raising A moves items the right way and never the wrong way. On the 72 shared informative items, going from $A = 1024$ to $A = 4096$ leaves all 60 decided items decided, and converts 2 of the 6 outvoted ones. The direction is what the theory demands; the magnitude rests on six items and should be read as such.

The budget rule

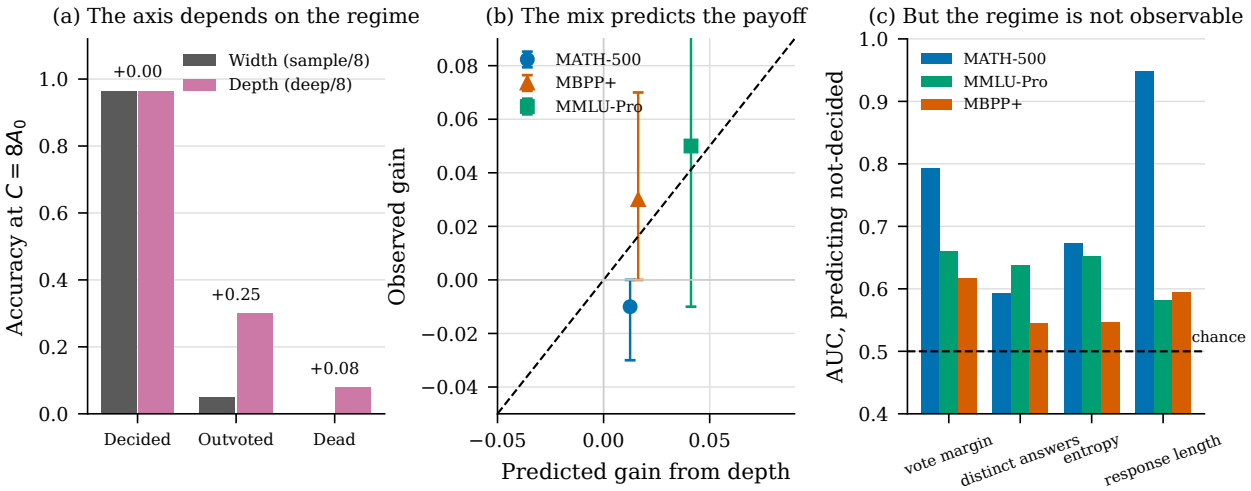


Figure 20: (a) At $C = 8A_0$, width against depth-in-one-call within each regime, with the difference annotated. (b) The gain predicted from the regime mix against the gain observed, with item-bootstrap intervals. (c) Whether a four-sample pilot can identify the regime; MATH-500 has only five non-decided items, so its bars are unstable.

Figure 20(a) measures the conditional choice directly. At $C = 8A_0$, spending the budget on one deep call rather than eight samples is worth

Regime	depth − width	95% interval
Decided (n=255)	0.000	$[-0.024,\ 0.024]$
Outvoted (n=20)	+0.250	$[0.050,\ 0.450]$
Dead (n=25)	+0.080	$[0.000,\ 0.200]$

which is exactly the pattern the theory predicts: no difference where a vote already works, a large difference where it cannot. Taking expectations over an item population with outvoted share O and dead share Z gives a rule for a whole problem type,

$$\Delta(\text{depth over width at } 8A_0) \;=\; 0.250\,O \;+\; 0.080\,Z.$$

Across the three informative benchmarks this predicts 0.013, 0.041 and 0.016 against observed values of -0.010, 0.050 and 0.030, and it matches the macro exactly at 0.023 predicted against 0.023 observed. Its largest error is 0.023, on MATH-500, whose outvoted mass is five items.

Why the rule cannot be conditioned on the prompt

The law conditions on (p, q_1), which is not known before sampling. The natural hope is to estimate it cheaply, spend a few samples, and route the budget per item. There is a structural reason this fails.

Observation. *The plurality margin measures consensus, not correctness.* Both p and q_1 are features of the same sampling distribution, and $\hat{m}$ estimated from k_0 draws converges to m at rate $k_0^{-1/2}$ regardless of whether the modal class is right. An item where the model is confidently correct and one where it is confidently wrong generate statistically similar samples: large $|\hat{m}|$, few distinct answers, low entropy. Nothing internal to the sampling distribution separates them, because the label is exactly what the distribution does not carry. Breaking the tie requires a signal from outside it: a verifier, an executable test, or a different model.

The data agree. Within benchmark, a four-sample pilot predicts that an item is not decided with an AUC of 0.66 on MMLU-Pro and 0.62 on MBPP+ using the vote margin, 0.64 and 0.54 using the number of distinct answers, and 0.65 and 0.55 using entropy. Pooled across benchmarks, response length looks informative at 0.73, but that is benchmark confounding: within MMLU-Pro and MBPP+ it falls to 0.58 and 0.60. The mean plurality margin on MMLU-Pro is 0.813 even though a quarter of its items are not decided.

We checked the consequence rather than assuming it. An adaptive policy that draws four samples, keeps the vote when the margin is wide and escalates to a deep call otherwise, reaches 0.850 at an expected cost of $8.19A_0$. That equals the best configuration in the entire study, `private_team/5/4` at $25A_0$, for a third of the compute, which is a real saving. But it sits only 0.003 above simply spending the same budget on `deep/8` with no gate at all. Conditioning bought the schedule, not the accuracy.

A worked example

Take a budget of $8A_0$ per item on MMLU-Pro, whose regime mix is 0.74 decided, 0.12 outvoted, 0.14 dead, and whose decided items have a mean margin near 0.9.

Spending it all on width gives eight samples. The decided items needed $k^*(0.9) = 1$, so seven of the eight draws are wasted on them; the outvoted items get worse, from 0.100 at four samples to 0.050 at eight; the dead items stay at zero. Delivered accuracy is 0.720.

Spending it on one deep call at $8A_0$ leaves the decided items where they were, lifts the outvoted ones from 0.050 to 0.300, and rescues 0.080 of the dead ones. Delivered accuracy is 0.770. The predicted gain, $0.250 \times 0.12 + 0.080 \times 0.14 = 0.041$, is close to the observed 0.050.

The reason the gain is not larger is that 0.74 of the budget is spent on items that were already settled after one sample. That is the deeper point: on this model and these benchmarks, most compute is spent on questions that were never in doubt.

Assumptions, and where they break

The derivation assumes draws are independent and identically distributed, which holds for the `sample` family by construction (independent roots) but not for revision or peer families, whose draws are conditioned on earlier ones. For those, Proposition 1 still describes the limit of their final answer distribution, but the variance in Proposition 2 understates the true one, since correlated draws carry less information than independent ones. This is visible as the elevated selection success of peer configurations in Figure 16: their draws are so correlated that the vote resolves almost immediately, on whatever answer they converged to.

The model assumes a plurality selector over answer classes, so it does not describe MBPP+, whose public-test selector scores candidates individually. That benchmark has 19.0 distinct answers among 25 samples and a top class holding 0.170, so the plurality abstraction is not merely inaccurate there but inapplicable, which is why it is excluded from the validation above and analysed separately in Section 2.13.

Finally, the normal approximation degrades at small k and near $m = 0$, where the discreteness of the counts and ties matter. The validation reports a mean absolute error of 0.007 overall but the error concentrates in the $|m| < 0.2$ band, which holds few items here and is exactly the band where the answer is that no feasible amount of width resolves the item anyway.

2.15 A design check

On all 400 items, a team of n agents run for zero revision rounds delivers exactly the same answer as n independent samples, under both the private and the peer arms. This is true by construction, since with no rounds there is nothing to revise and nothing to see, and its exact agreement (400 of 400 items for every n) confirms that the iso-compute re-keying used throughout this report lines up with the sealed selection records.

3 Interpretation

At fixed compute, the shape of the allocation is close to irrelevant. This is the least exciting reading of the data and the best supported one. Seventy paired comparisons per benchmark produce no effect that survives correction, and the spread across all allocations at the largest budget is about 5 accuracy points on the macro, comparable to the width of a single confidence interval. Claims that multi-agent structure beats

single-agent inference should be checked against a compute-matched sampling baseline, because on this model and these benchmarks that baseline is already most of the way to the best result anyone achieves.

Test-time compute keeps working; test-time *selection* does not. Oracle pass@k rises steadily and without any sign of saturating below 0.92 on the macro, which means additional compute is still producing correct answers. What fails is the readout. Plurality voting stops improving after roughly two samples, and on MBPP+ the public-test selector gets actively worse as the pool grows, because more candidates means more candidates that satisfy the visible evidence and fail the hidden suite. The gap between what was generated and what was delivered is the largest single quantity in this study, and it grows with every dollar spent.

Communication between agents is a cost, not a benefit, in this regime. The prespecified contrast is negative and significant, the effect deepens monotonically with rounds, and the round dynamics show why: peer exposure increases the rate at which correct answers become wrong more than it increases the reverse. A small model shown its peers' drafts converges, and convergence on a wrong answer is indistinguishable at readout time from convergence on a right one. The private team, which gets the same number of agents and rounds with no mutual visibility, dominates it everywhere. This should be read as a statement about small models with a plurality readout, not about multi-agent debate in general, but it does mean the burden of proof runs the other way.

Width beats depth slightly, and one deeper call beats both on cost. Among private allocations, spreading a budget over three to five agents edges out spending it on one agent revising many times, though the margin is inside the noise. More striking is that a single call with the whole budget as reasoning allowance matches the best orchestrated configuration on the macro to within 0.3 points and wins outright on MBPP+, while requiring no orchestration, no repeated context and no selector. If a practitioner has to pick one thing to do with extra tokens on this model, the evidence points at that.

A prescription, in order of what it is worth. The levers can be ranked on this data, at $C = 25A_0$ on the three-benchmark macro.

First, verification. Moving the selector from its current effective AUC of 0.63 to 0.80 is worth 4.3 points, from 0.833 to 0.876, on candidates already generated. Even AUC 0.70 beats every allocation in the study. Nothing else on this list comes close, and it is the only lever that scales with the compute already being spent, because a better selector makes every additional sample worth something.

Second, per-draw reasoning. Raising A is the only generation-side axis that touches the binding quantity, and it works where longer reasoning raises per-draw accuracy: it halved the MATH-500 gap, did nothing on MMLU-Pro and hurt on MBPP+. Verify per benchmark before spending on it.

Third, fix broken selectors. MBPP+ public-test selection is below random from $k = 8$. This is not a research problem, it is a defect, and repairing it is worth 1.2 points immediately and unlocks the 10-point coverage headroom that sits behind it.

Third-and-a-half, adjudicate the top two where the answer space is narrow. On MMLU-Pro one binary comparison per item is worth $+0.080$ $[0.030, 0.140]$, two thirds of the gap, and needs a single extra call rather than a full reranking pass. This does not transfer to code, where the correct program ranks 7.43 among failures and no shortlist is short enough.

Fourth, consolidate rather than communicate. If the readout is a vote, prefer allocations that raise the correct share of the pool. Private teams do this and are worth 1.7 points over sampling; peer exposure does the opposite. Read out over the archive rather than the last round, which is free and recovers 3.7 points of the damage peer exposure causes.

Fifth, stop buying width past k^*. Width resolves a decided item in $k^* \approx 1.04\, m^{-2.17}$ samples, which is one or two for most of the panel, and does nothing after that. On outvoted items it is worse than nothing, since the vote converges on the wrong answer. Depth is the only axis that changes the regime, and at $C = 8A_0$ it is worth $+0.250$ $[0.050, 0.450]$ there.

Last, and close to worthless, more draws. At fixed A with a vote as the readout, additional agents or rounds move delivered accuracy by about nothing. They raise the ceiling and leave the floor where it is.

The one-line version: this model does not need to think more times, it needs someone to tell it which of its answers was right.

A A validation experiment for the prescription

The prescription rests on a simulation: a reranker of given pairwise AUC, applied to the pool compositions this run actually produced. It bounds what discriminative power is worth. It does not show that a verifier with that power is available for these tasks. The following design tests that, and it is cheap because the candidates already exist.

Hypothesis. A judge model scoring stored candidates delivers higher accuracy at $k = 25$ than the plurality vote, by at least the 4.3 macro points the simulation predicts for AUC 0.80.

Design. No generation. The 25-sample bank for each of the 400 main-panel items is already stored, graded and sealed, so the run scores existing candidates and re-reads the outcome. Arms:

1. *Pointwise judge*, Flash-Lite. One call per candidate scoring correctness on a fixed rubric without the reference answer. 10,000 calls.
2. *Pointwise judge*, Flash, as a capability check on the same items.
3. *Top-two adjudication*, one pairwise call per item between the two most-voted answers. On the analysis above this is the highest-value arm on MMLU-Pro and the cheapest to run, at 400 calls rather than 10,000. It also tests the hard part directly, since the adjudicator must be willing to overrule a majority whose mean margin is 0.813.
4. *MBPP+ selector repair*, at zero API cost: re-select among stored candidates that pass the public assertion, breaking ties by judge score rather than by order.

Primary outcome. Delivered accuracy at $k = 25$, macro over the three informative benchmarks, paired per item against the sealed plurality readout. **Secondary.** The judge's realised pairwise AUC, which places it on Figure 14 and tests the simulation itself rather than only its conclusion.

Power. The median standard error of a paired macro contrast on this panel is 0.0119 at 400 items, so the minimum detectable effect at 80% power is 0.0333. The predicted effect of 0.043 clears it; a 192-item panel, with a detectable effect of 0.0481, would not. Use the full 400 items.

Cost. At roughly 1082 completion tokens per stored candidate, the pointwise arm is about 16.8M input and 1.28M output tokens: $2.19 on Flash-Lite and $8.25 on Flash. The pairwise arm adds ten comparisons per item. The whole study is comfortably inside $25, against $47.64 already spent of a $135 ledger ceiling. The shared account balance should be checked before launching, since it is not reserved for this study.

Prespecify before running. The rubric and its version hash, the scoring temperature, the tie-break rule, the exclusion of the reference answer from the judge context, and the two primary comparisons (pointwise judge against plurality, and pairwise against pointwise) with Holm correction over exactly those two. The judge must never see the grader output; the existing seal-before-join discipline applies unchanged.

What would falsify the prescription. If a competent judge lands at or below AUC 0.63, which is what plurality already achieves, the selection headroom identified here is not reachable with an LLM judge on these tasks, and the conclusion narrows from "spend on verification" to "this particular verifier does not work", with the gap remaining open.